Introduction to MTB Suspension Setup & Handling

The following is an attempt to enlighten people who might be relatively new to mountain biking. Bear in mind that there is no ideal, universal setup. Suspension is 100% personal, and what works for someone else might be awful for you. What feels great in the carpark might make you crash on a tricky bit of trail. Be prepared to learn, unlearn, and relearn everything you think you know about your bike setup preferences!

Please note that it's not worth trying to put any of this into action if you have suspension that is in desperate need of servicing or rebuilding.

Maybe you've just invested in your first dual suspension bike and truly have no idea what you should do with all those adjustments. This is not an exhaustive analysis, nor is it meant to be. I'm always eager to talk to people about their riding, their setup and look for ways to improve bike handling.

Conversely, there's a lot of riders out there who have been on MTBs for years, but always maintained a 'set & forget' mentality and never bothered experimenting with settings. Hopefully you guys get something out of this as well!

Here is my take on suspension setup, common misconceptions and potential ways for you to get the most out of your setup and handling. Everything you read here is my opinion - based on hundreds of hours of research, experimenting with different setups and learning about bike handling – as well as a sound understanding of the inner workings of springs and dampers on mountain bikes. It's also worth remembering that Honda's MotoGP team still haven't found the perfect suspension setup for smooth tarmac race tracks. What hope is there for mountain biking?? 😊



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First – let's talk about handling, comfort and control. It's helpful to know whether you're trying to improve handling as you explore the outer limits of your courage, trying to spend more time on the bike without getting your hands, feet and butt beat up, or trying to improve your ability to control the bike and make it do exactly what you want it to do.

In my universe, "handling" refers to the behaviour of your bike only when pushing up to or beyond the bikes limits for your particular setup. No matter how good a rider you are, tyres only have finite grip. Suspension only has limited response. Frames have specific flex traits and geometry range. Handling refers to what the BIKE does, when you approach terrain with a speed and attitude that is outside what you consider a comfortable and safe speed.

"Comfort" is pretty literal. How safe do you feel when riding within your limits? All the small bumps get soaked up, landings aren't overly harsh, your hands and feet don't get beaten up during a normal ride.

Control is a crossover between your ability to make the bike do what you want, and the bikes ability to meet those expectations and do what you're expecting it to do. I suppose you could substitute 'control' for 'trust' to a certain degree. You can maintain your weight over the bars in a rooty, off camber corner because you TRUST that you won't lose the front, if you maintain your intended line, or that if you wander off your chosen line, it will react in a way that you can comfortably control because you trust your setup. You trust that you won't get kicked over the bars if your rear tyre encounters a big hit that you didn't see or couldn't miss. You can control the bike down steep chutes where you're at the limit of mechanical traction and your bikes geometry.

It's worth noting that while control and handling go hand in hand, control and comfort are mutually exclusive. A bike that is comfortable and squishy around your favourite trails at your normal speed will likely be outrageously hard to control the moment you try and ride anything fast, steep or rough (or all 3!)



Photo Courtesy of Dan Hearn @ VitalMTB

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Starting with Sag

Static Sag is the amount that your suspension settles into its travel under the weight of you, your gear and the bike. It is a major determinant of your bikes ride height, geometry, and it's 'attitude' (i.e poppy/easy to jump or squishy/hard to jump). Sag also determines how much capacity your wheels have to track the ground as it moves away from your tyres.

Here's how to find your initial sag setting:

1) Read your owners manual to find out what their recommended sag point is.
   1a) Find out what your shock stroke dimension is (how far the shaft moves) If, for example, your stroke length is 60mm, then 30% rear sag means the o-ring sits 18mm down from the air can seal
   1b) Same rules apply to forks. If you run a 140mm travel fork, then 20% sag means the o-ring sits 28mm above the fork seals.
2) Wear all your riding gear and carry all spares/water you'd normally carry.
3) If you have a SRAM rear derailleur, lock it in the forward position to make the chain completely slack, or just use your quick-link to remove the chain completely from a Shimano equipped bike.
4) Set all compression adjustments to their open positions, set all rebound adjustments wide open (fast).
5) Lean the outer edge of your hand grip against something for stability but rely on your own balance as much as possible – then get into an aggressive stance with most of your weight on your feet.
6) Bounce the bike a few times, get back into your attack position and very carefully reach down to your sag o-rings and push them back towards the fork/shock seals without compressing the bike any further.
7) Gently dismount, check your sag o-rings, and add/remove pressure to get the number you need - lets call it 30% sag rear, 20% front.

Step 1 is just a starting point. For ultra-steep trails you might run more sag. For all day epics grinding up and down fire roads you might run less sag.

Step 3 is where my method deviates from most others. All modern 1x drivetrain derailleurs have some kind of clutch mechanism to manage chain slap. This clutch mechanism acts as a friction damper and makes it harder for your bike to move through its travel, giving you inaccurate sag numbers.

Something else to consider as you progress with the sport is Dynamic Sag – the average sag % of your suspension whilst you're actually on the trail. I'll cover this in more detail at a later date.

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Tyre pressure

Your tyres are the first line of defence in bump absorption and traction and are a critical component of the suspension system. There is no point in running a tyre pressure so low that you can bottom the tyre out on the rim before the fork or shock bottoms out. The tyres must be able to support the force of a full suspension compression.

I've found that a difference of 2-3 PSI can transform my bikes comfort and handling from 'pretty terrible' to 'oh my god that feels awesome' on different trails. No right or wrong, only personal preference – with the caveat that you need to at least run enough pressure NOT to pinch flat, damage your rims or break your tubeless tyre/rim seal (called 'burping') under hard cornering.

I'm a heavier guy – 95kg fully loaded with water and gear. I've found that to at least get in the ball park of my ideal pressure I need to be able to apply maximum braking force to the front tyre on a steep trail, without the tyre starting to wrinkle under the strain. I tend to only bother with 2 sets of pressure. 22-24PSI (front to rear) for trails built in soil and roots, and 25-27PSI for fast rocky trails – just to give me some extra protection against rim strikes.

If I know that my front tyre still has some support left in reserve even under maximum braking effort, then it should have the ability to soak up any roots/rocks I hit whilst slowing down.

There are a lot of factors that determine what your optimal tyre pressure might be. Rim and tyre width, whether you run tubes or tubeless, what you weigh and how aggressively you ride…

Now – you need to record your pressures and go test them on the trail. Of course, if you ever feel you're at risk of pinching tyres or impacting your rims on the trails you ride, these pressures have to increase. Conversely, you can probably get away with lower pressures on slow, technical trails than on high speed rough trails. I would encourage you to get a cheap (sub $50) tyre pressure gauge to help you be consistent and accurate.



Photo Courtesy of Trev Worsey @ Enduro MTB

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Rebound Settings (initial)

Rebound is the speed at which your suspension recovers from a compression. Rebound adjusters are normally red. Rebound drastically affects handling and control in rough terrain, as it determines how much of your suspension travel is available to take the next impact.

In the context of handling and control traits, if you've ever seen a video of someone attempting a jump and ending up being bucked forward on take-off (to the inevitable front-heavy landing/crash) – you've probably witnessed someone who is running the rebound on their rear shock way too fast (or, someone who just sucks at jumps)

If you're just starting to learn to do real jumps, sometimes you can gain some confidence by slowing your rear shock rebound down 2-3 clicks (assuming you already have reasonable jumping technique)

So – where to begin? Well, most newer generation forks are now being supplied with recommended pressure and rebound settings based on your weight. A rebound setting is a number of clicks out from the fully closed (very slow recovery) position. Heavier riders running more pressure need to run fewer clicks out from closed due to running a higher spring rate (there is more energy for the rebound circuit to dissipate). Lighter riders running less pressure can run more clicks out from closed owing to there being less stored energy in the air spring.

It's worth noting that SRAM have recently released the TrailHead app to help riders setup up RockShox suspension products. Go to www.trailhead.rockshox.com/en/ to check it out.

If you have NO idea where to start, here's a very basic starting point for your fork and shock:

Assume rebound adjuster has a range of 20 clicks		
Rider Weight	**Clicks out from fully closed - Fork**	**Clicks out from fully closed - Shock**
45kg	12	10
45-55kg	11	9
55-65kg	10	8
65-75kg	9	7
75-85kg	8	6
85-95kg	7	5
95-105kg	6	4

Note that there are still a number of clicks remaining at the extreme ends of adjustment, and that the shock is generally going to be running more rebound damping (slower) than the fork.

This table does not take into account your ability, the trails you're riding or how aggressively you ride. It's only to serve as a reasonably safe place to start for those who are completely at a loss as to where they should begin. If your suspension has a different number of clicks, adjust your mid point accordingly and click higher/lower from there.

Experiment with different rebound settings and find out what works for you.

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Low Speed Compression (LSC)

Low speed compression, which I'll henceforth refer to as LSC, refers to the speed of the damping shaft moving through the fork/shock body – NOT your actual riding speed. Think of LSC events as gentle bobbing as you pedal around, or the compression of the fork and shock as you load the bike up through a nice high berm. These are LSC events – slow shaft speeds. Your LSC adjuster affects how these low speed shaft movements are handled by your suspension.

I like to say that LSC changes are 'experienced' rather than 'felt', especially when you adjust only one click at a time. LSC is perhaps the most frustrating adjustment for riders as a single click may be indistinguishable from the last in the carpark, but be the magical Goldilocks setting up on the trail

LSC sets up the attitude of the bike under hard braking and cornering by doing two main things:

1) Increased LSC damping on your fork allows it to support you better under hard braking on a steep trail. This means that the geometry of your bike is better preserved, and helps to keep you from going over the bars (OTB!) Having more travel in reserve also means the fork is better prepared to soak up any bumps in your way, whilst you're hard on the brakes.
2) LSC damping is also an effective way to load up the OPPOSITE tyre. This blows peoples minds when they first hear it – but it's just Newtons 3rd Law of motion. For every action there is an equal and opposite reaction. Increasing the LSC of the fork means the equivalent increase in support is transferred to the rear tyre via the fulcrum (the bike frame).

Consider that second point for a moment – if you5 find your front tyre is a bit nervous, you can gain traction by adding LSC to your rear shock. For most people that simply means going from the open setting to the mid setting. The increased damping force at the rear wheel loads up the front tyre to a greater extent, experienced as an increase in front wheel grip. Where your front wheel might have tried to let go before, you'll find you can now really lean on it through turns (up to the limit of mechanical grip or available talent!)

Something most people have used is the climb switch on their rear shock – and you probably noticed straightaway that the back of the bike rode higher in it's travel and you were less prone to pedal strikes. That's another example of LSC in action.

Experiment with small LSC adjustments on trails you're familiar with.



Photo Courtesy of BIKE Magazine

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Suspension Progressivity & Volume Spacers

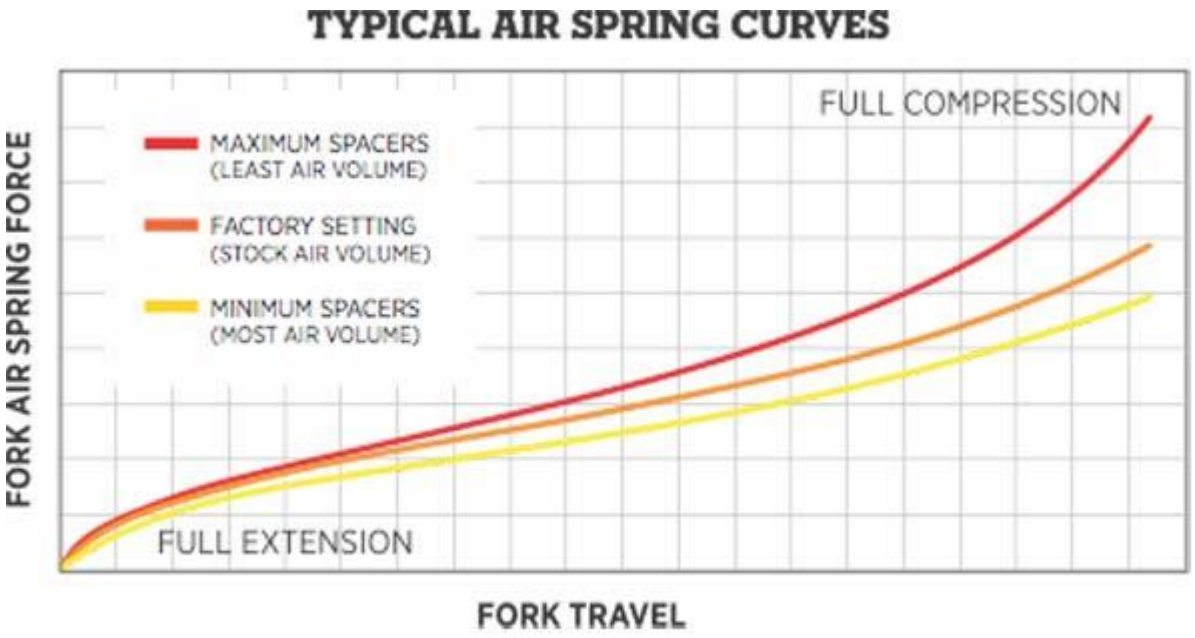
If you're new to the world of MTB, it's unlikely you'll need to play with the volume of your air spring unless you're exceptionally light or heavy but It's a concept worth understanding to help you as you get faster.

> *Adding spacers to your suspension reduces the overall volume and makes the spring curve more progressive –* ***more supportive beyond sag point and harder to bottom out.***
>
> *Removing spacers from your suspension increases the overall volume and makes the spring curve more linear –* ***less ramp up beyond sag point, easier to bottom out.***

Air springs can be linear, progressive, digressive or some combination of all 3 but that's way beyond what we're talking about for a beginners guide.

Adding volume spacers to your air spring reduces the overall volume of the chamber, thus making the spring rate ramp up deeper into the travel. Some riders have issues where they might want to run 30% sag but blow through the travel way too easily. Other riders have the exact opposite problem. Adding or removing spacers might be the solution you're looking for.

Check out the graphic from Fox Racing Shocks below and notice that the spring rate around sag point (25-30% of the way along the horizontal axis) is pretty similar for all three setups – the full effect of the volume reduction isn't felt until the very end of the stroke.



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Let's wrap this up…

All of this theory aside, it's important to remember that there is no universally ideal setting – only that which suits you best in the exact moment you need it. Back when I did my factory training with SRAM, I was lucky enough to check out both Sam Hill and Troy Brosnan's bike setups from their time with the Specialized team. Troy had softer spring rates and rebound so slow I thought someone had tampered with it. Sam had an absurdly stiff setup with rebound so fast I couldn't possibly imagine trying to ride that bucking bronco. Some people think that if pros are running stiff/firm setups, that must be the key to going faster. This is fine, so long as you have the physical strength to support yourself on a bike that requires 4-5x your body weight to make it bottom out! Horses for courses, everyone has a different style and preference.

If you're lucky enough to be of the "set and forget" mentality, suspension settings will likely never trouble your riding or keep you awake at night. However, if you're really wanting to progress with your sport, seeking out higher speeds, more exciting thrills and longer dances with the elusive, intoxicating feeling that is "flow" – then some educated suspension adjusting (and annual servicing!!!) will help you get there.

Don't be afraid to try the exact opposite adjustment to what everyone tells you to try (that includes anything I've written above!), as long as you're able to manage the consequences of it turning out to be an awful adjustment. In light the of that, here's the final handy chart for you to peruse and maybe even try out some of the suggestions. Treat changes A & B like "and/or" options. Pick either, both, or neither. It's up to you.

Terrain/Speed/Handling Problem	Suggested Change A	Suggested Change B
Flowy, fast & bermed	Less sag front and rear	Firmer LSC front and rear
Steep, rocky and slow	More sag in the rear	Firmer LSC on the fork
Fast and rocky	Less sag front and rear	Faster rebound front and rear
Front tyre washout	Less tyre pressure front	Less rear sag
Rear tyre washout	Less tyre pressure rear	Firmer LSC on the fork
Tyres squirming under high loads	More tyre pressure front and rear	Ride slower? Nahhhhh!
Kicking forward on jump take-off	Slower rebound on the rear shock	Faster rebound on the fork
Mud, roots, slippery everywhere	Wide open LSC front and rear	Slower rebound front and rear
Packing down over rough sections	Faster rebound front and rear	Firmer LSC front and rear
Too much fork dive under braking	Firmer LSC on the fork	Less sag in the fork
Pedal strikes	Less sag in the rear	Firmer LSC in the rear shock
Higher speeds than normal	Firmer LSC front and rear	Faster rebound front and rear
Lower speeds than normal	Less LSC front and rear	Slower rebound front and rear

…Now go and ride!

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The Local Spokesman is a mobile bicycle mechanic offering a high level of service tailored to road and MTB riders in Perth, Western Australia.

TLS is owned and operated by Kyle Walter, a lifelong bike fanatic and perpetual fiddler.

Check us out online by clicking any of the images below.



